

इस हाशिए
में केवल
प्रश्न संख्या
लिखें।

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Specimen Booklet

UPSC

For Practice Purpose Only

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चाहिए।

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63th
BPSC

Ans 1(a)

$$\text{Total units sold by company X in 2013} = \frac{3.5 \text{ L}}{7}$$

$$= \frac{350000}{7}$$

$$= 50000 \text{ units}$$

$$\text{Total units sold by company Y in 2013} = \frac{4.5 \text{ L}}{5}$$

$$= \frac{450000}{5}$$

$$= 90000 \text{ units}$$

$$\text{Total units sold by company Z in 2013} = \frac{4 \text{ lakh}}{8}$$

$$= \frac{400000}{8}$$

$$= 50000 \text{ units}$$

$$\text{Average price of soap in year 2013} = \frac{\text{Total price}}{\text{Total units sold}}$$

$$= \frac{3.5 \text{ L} + 4.5 \text{ L} + 4 \text{ L}}{50000 + 90000 + 50000}$$

$$= \frac{1200000}{190000}$$

$$= ₹ 6.31$$

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(b) units sold by company X in 2011

$$\frac{300000}{5}$$

$$= 60000 \text{ units}$$

units sold by company X in 2012

$$\frac{300000}{6}$$

$$= 50000 \text{ units}$$

Percentage decrease in units sold for X in

2012 over 2011 = $\frac{\text{units sold in 2011} - \text{units sold in 2012}}{\text{units sold in 2011}} \times 100$

$$= \frac{60000 - 50000}{60000} \times 100\%$$

$$= \frac{10000}{60000} \times 100\%$$

$$= 16.66\%$$

(c) Units sold by company X in 2013

$$\frac{350000}{7}$$

$$= 50000$$

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$$\begin{aligned} \text{units sold by company Y in 2013} \\ &= \frac{450000}{5} \\ &= 90000 \end{aligned}$$

$$\begin{aligned} \text{units sold by company X in 2013} \\ &= \frac{400000}{8} \\ &= 50000 \end{aligned}$$

$$\begin{aligned} \text{Total units sold} &= \text{units sold by X} + \text{Y} + \text{Z} \\ &= 50000 + 90000 + 50000 \\ &= 190000 \end{aligned}$$

$$\begin{aligned} \text{Percentage share of Z in total units sold} \\ \text{in 2013} &= \frac{\text{share of Z}}{\text{total units sold}} \times 100\% \\ &= \frac{50000}{190000} \times 100\% \\ &= 26.31\% \end{aligned}$$

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Year	Units sold by Y
2011 ✓	$\frac{440000}{4} = 110000$
2012	$\frac{420000}{6} = 70000$
2013	$\frac{450000}{5} = 90000$
2014	$\frac{490000}{7} = 70000$
2015	$\frac{420000}{6} = 70000$

The maximum no. of units sold by Y in year 2011 i.e., 110000 units.

(e) Maximum price per unit given in table = 9
Minimum price per unit given in table = 4

$$\text{Required percentage} = \frac{\text{Max price} - \text{Min price}}{\text{Min price}} \times 100\%$$

$$= \frac{9 - 4}{4} \times 100$$

$$= \frac{5}{4} \times 100$$

$$= 125\%$$